

SAQIBALI

ELECTRICAL ENGINEERING STUDENT

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PROFILE

A driven engineering student with a solid grounding in Real-time Control Systems and Signal Processing. Experienced in creating adaptive control solutions for high-inertia systems. Proficient in Fusion 360 for 3D modelling and design, and skilled in conducting Finite Element Analysis (FEA) to evaluate product performance and durability. Enthusiastic about emerging technologies, such as Generative AI and aerospace applications, and eager to apply technical skills in a challenging engineering position.

EDUCATION

Zakir Husain College of Engineering & Technology (ZHCET), Aligarh

B.Tech in Electrical Engineering

Sep 2023 - Jun 2027

Syedna Tahir Saifuddin (STS) School, AMU

Senior Secondary School (PCM)

PROFESSIONAL EXPERIENCE

Research Intern

Indian Institute of Technology, Ropar | Rupnagar | June 2026 - July 2026

- Developed mathematical models and designed Field-Oriented Control (FOC) algorithms for a Five-Phase IPMSM.
- Generated ePWM signals via TI DSP using Code Composer Studio (CCS) and tested the gate driver circuit.

Lead Engineer

Orbitalink Private Limited | Aligarh | August 2025 - Present

- Led the end-to-end development and integration of advanced antenna tracking systems, bridging hardware design with firmware programming.
- Conceptualized a custom 435 MHz Yagi-Uda antenna and designed the mechanical tracker chassis using Fusion 360.
- Developed Arduino-based firmware to control hybrid servo motors via HSD57 drivers.
- Managed the field unit's physical infrastructure and cable routing while troubleshooting complex hardware-software bottlenecks to ensure seamless system operation.

Electrical Design Intern at Orbitalink Private Limited (in collaboration with AMU Innovation Foundation) | Aligarh, UP

- Developed and implemented an Adaptive PID Controller designed to achieve stabilization and precise control of a high-inertia antenna tracking system.
- Applied advanced control theory principles, system modelling, and performance optimization for critical satellite communication infrastructure.

Ground station Team Member

SS AMU SAT | Aligarh, UP

- Played a key role in developing and building the 3D model chassis for the antenna tracker.
- Conducted tests on antenna tracking mechanisms.
- Assisted with ground station operations and satellite tracking protocols.

- Helped maintain and calibrate communication equipment to guarantee reliable data transmission.
- Acquired hands-on experience in signal reception, hardware troubleshooting, and system diagnostics.

SKILLS

Hard Skills & Expertise

- **Development Skills:** Proficient in Python, Full Stack Web Development, and Data Science (including NumPy, Pandas, Matplotlib, and Seaborn), as well as sensor interfacing and I2C protocols, with expertise in C++ for hardware control.
- **Engineering Expertise:** Skilled in Mechanical Product Design utilizing FUSION 360, Control Block Design, and Simulink, with additional expertise in Field-Oriented Control (FOC) and Mathematical Modelling.

Soft

- Communication
- Problem Solving
- Team Collaboration
- Leadership

PROJECTS

Weather Dashboard Application

[Live Deployment](#) \ [Source Code \(GitHub\)](#) Open Weather Map API, Node.js, Axios

- Deployed on Render using Open Weather Map API, Node.js, and Axios.
- Developed a responsive weather forecasting app with Modern JavaScript and CSS3.
- Integrated Open Weather Map API for real-time data on temperature, humidity, and wind speed globally.
- Created a robust search function for querying weather conditions worldwide.
- Established error handling protocols to manage API failures and invalid inputs.

Simulink PID Control System Modelling

File: [PID_SIMULATIONV1.slx](#) | Platform: Simulink

- Designed and simulated a dynamic control loop in Simulink to model the behaviour of a physical plant.
- Verified controller performance against various setpoint changes and noise inputs.

Simulink Modelling and Analysis of a DC-DC Boost Converter

File: [Model Boost.slx](#) | Platform: Simulink

- Designed and simulated a DC-DC Boost Converter circuit in Simulink as part of college laboratory work, showcasing the ability to step up a DC input voltage.
- Analysed the converter's steady-state and transient responses under varying duty cycles and load conditions.
- This project demonstrates proficiency in Power Electronics, Circuit Simulation, and Switch-Mode Power Supply (SMPS) fundamentals.

3D Chassis Design for Motorized Tracking Unit

- Crafted the 3D model of the chassis for the motorized tracking unit using Fusion 360, optimizing the design for rigidity and minimal weight while ensuring proper placement for the motor, gearbox, and sensors.
- Implemented the core PID control logic for the two-axis antenna positioning system, which is crucial for maintaining a lock on a satellite signal.

CERTIFICATIONS

Certificate of Participation in MATLAB Simulation and Control

IEEE Student Branch, AMU

Advanced Aircraft Design Workshop

Bharat Space Education Research Centre

Certificate of Participation

SS AMU SAT

Data Science Course

Certificate of Internship - Explore Electrical Engineering Job Simulation

GE Aerospace

ACTIVITIES

SSAMUSAT Antenna Tracker Development Team

Position Held: Member of the Ground Station Team

Tenure: September 2024 - Present

Role Description: As an enthusiastic participant in the university's satellite development team, I am committed to the meticulous design and construction of a sophisticated ground station antenna tracker, ensuring high precision and reliability in satellite communication.

- Implemented the core PID control logic for the two-axis antenna positioning system, which is crucial for maintaining a lock on a satellite signal.
- Advanced Aircraft Design Workshop (2024): Participated in a comprehensive workshop focused on the intricacies of aircraft design, enhancing both theoretical understanding and practical skills.
- SS AMU SAT Certificate of Participation (2024): Received certification for active involvement in the workshop conducted by SS AMU SAT project.

IEEE Student Branch, AMU

- Attained a certificate for participation in Matlab Simulation and Control (Issued Oct 2025).
 - Completed Introduction to Cybersecurity, acquiring fundamental knowledge in Cybersecurity.
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